

T2 Coursework Report (ELEC70073 Computer Vision & Pattern Recognition)

Anson Chin (CID: 02194736), Samuel Khoo (CID: 02264291)

Introduction

This report examines whether object shape and material can be identified using tactile sensing alone. Data were collected using a UR5e robot equipped with a Contactile tactile sensor and wrist-mounted force/torque sensor during repeated controlled contacts with objects of different geometries and materials. All processing and pattern recognition methods are implemented in the accompanying MATLAB code. Colour labels for shapes are provided in the Appendix.

A. Data Preparation and Visualisation

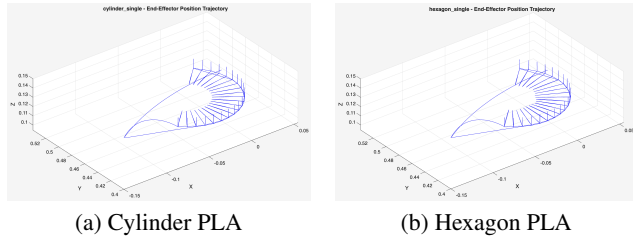


Figure 1: [A.1.a] End-effector position trajectories for cylinder and hexagon PLA objects.

[A.1.b] The cylinder trajectory is smooth and rotationally symmetric, reflecting uniform surface normals during contact. In contrast, the hexagon trajectory shows subtle deviations associated with edge- and face-dependent interactions, leading to less uniform stopping positions across trials.

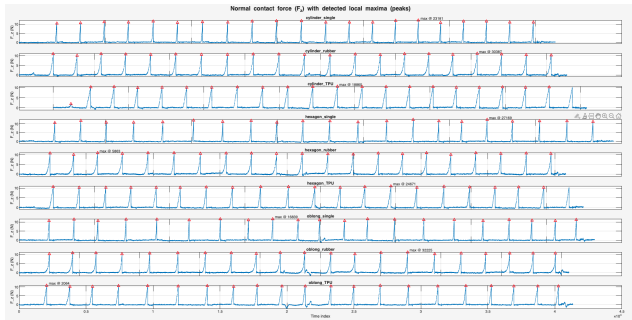


Figure 2: [A.2.a] F_z time series for all objects with detected peak contact indices marked.

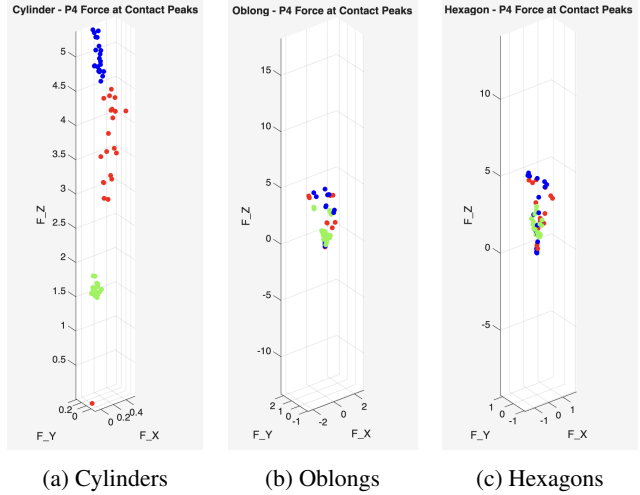


Figure 3: [A.3.a] 3D force scatter plots from the central papilla (P4) at peak contact force for cylinder, oblong, and hexagon objects.

[A.3.b] **Observations** Across all shapes, stiffer PLA objects produce tighter force clusters, while compliant rubber shows greater spread due to increased shear deformation. Shape effects are most evident for oblong and hexagon objects, where orientation and edge contact introduce additional variability in force direction and magnitude.

B. Principal Component Analysis

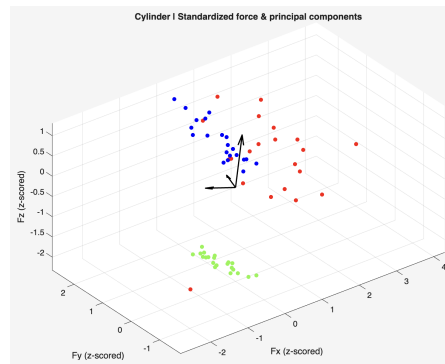
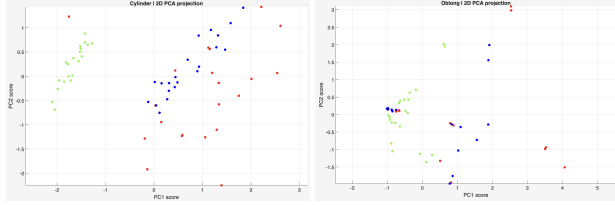
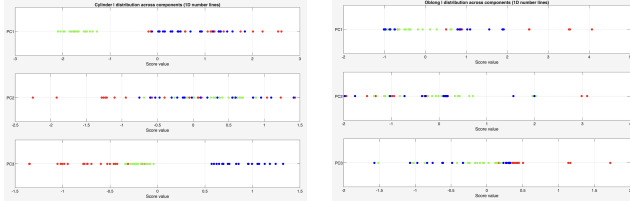


Figure 4: [B.1.a] Cylinder – Standardised data with PCs



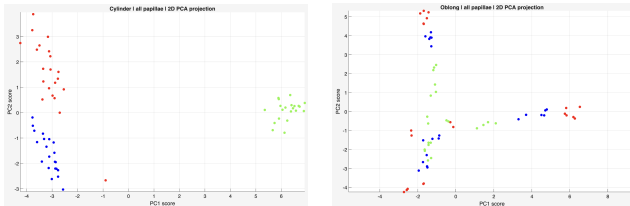
(a) [B.1.b] Cylinder – 2D PCA (PC1 vs PC2) (b) [B.2.a] Oblong – 2D PCA (PC1 vs PC2)



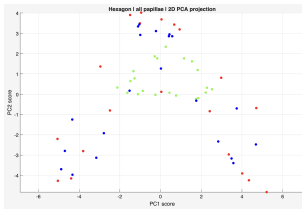
(c) [B.1.c] Cylinder – 1D PC distributions (d) [B.2.a] Oblong – 1D PC distributions

Figure 5: Principal Component Analysis of P4 force data for cylinder and oblong objects.

[B.2.b] Cylinder vs Oblong Comparison Compared to cylinders, oblong objects exhibit greater variance along the leading principal components, reflecting increased orientation sensitivity during contact. This results in less compact clustering and increased overlap between materials in the PCA space.



(a) Cylinder (b) Oblong



(c) Hexagon

Figure 6: [B.3.a] 2D PCA projections using all papillae force data for cylinder, oblong, and hexagon objects.

[B.3.b] Scree Plots and Interpretation The scree plots show that the majority of variance is captured by the first few principal components, indicating strong correlations and redundancy across papillae. Oblong objects exhibit a

slower variance decay, suggesting higher effective dimensionality due to orientation-dependent contact effects.

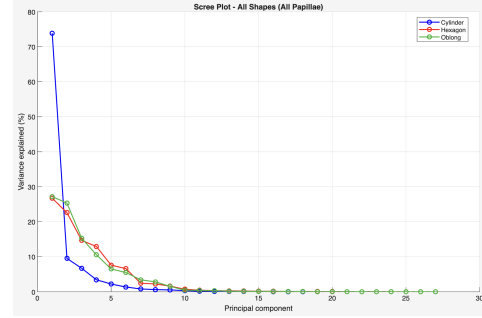


Figure 7: [B.3.b] Scree plot showing explained variance across principal components for all objects.

C. t-SNE

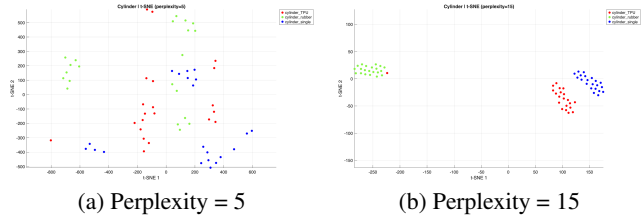


Figure 8: [C.1.a] t-SNE 2D embeddings of cylinder P4 force data for two perplexity values. Final KL-divergence losses are 0.8957 (perplexity 5) and 0.145 (perplexity 15).

[C.1.a] t-SNE on Cylinder P4 Force For cylinder objects, t-SNE was applied to the standardised P4 force data using two perplexity values. Perplexity 15 achieved a much lower loss than perplexity 5, indicating a more stable embedding that better preserves local neighbourhood structure.

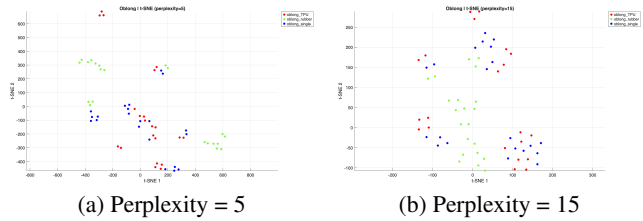


Figure 9: [C.1.b] t-SNE embeddings of oblong P4 force data using the same perplexity values as the cylinder experiments. Final losses are 0.3266 (perplexity 5) and 0.2287 (perplexity 15).

[C.1.b] t-SNE on Oblong P4 Force The same perplexity values were applied to the oblong objects to allow direct

comparison with cylinders. In contrast to cylinders, the oblong embeddings show less sensitivity to perplexity choice, with comparable loss values across both settings.

[C.1.c] Outcome Comparison Across Objects Cylinder objects exhibit more compact and well-separated clusters under higher perplexity, while oblong objects show greater dispersion due to orientation-dependent contact and increased variability in force patterns.

[C.1.d] Comparison to PCA Unlike PCA, which preserves global variance structure, t-SNE emphasises local neighbourhood relationships and reveals clearer material-based clustering. However, t-SNE embeddings are less interpretable in terms of physical force directions and are sensitive to hyperparameter choice.

D. Linear Discriminant Analysis

[D.1.b] 3D Displacement Scatter Plot Figure 10 shows a 3D scatter plot of the tactile displacement for the two materials. The distributions exhibit substantial overlap, suggesting limited separability in the raw displacement space.

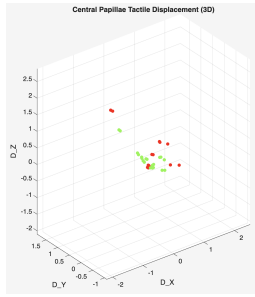


Figure 10: **[D.1.b]** 3D scatter of central papillae displacement for oblong TPU and rubber objects.

[D.1.c] Pairwise LDA on Displacement Components Linear Discriminant Analysis was applied to all pairwise combinations of displacement components: (D_X, D_Y) , (D_X, D_Z) , and (D_Y, D_Z) . The resulting projections and decision boundaries are shown in Figure 12b. Classification accuracies of 71.43% were achieved for (D_X, D_Y) and 90.48% for (D_X, D_Z) , while (D_Y, D_Z) achieved a lower accuracy of 54.76%.

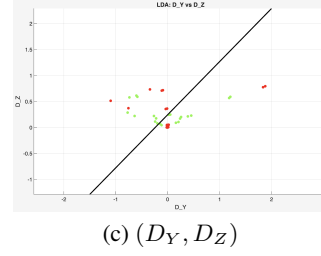
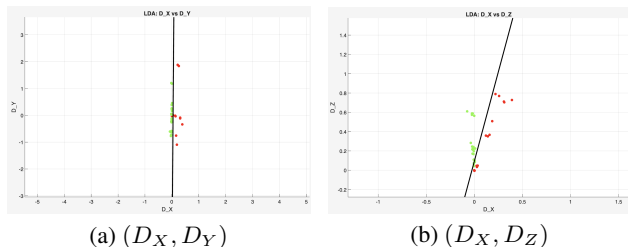


Figure 11: **[D.1.c]** Pairwise LDA results (continued).

[D.1.d] LDA on 3D Displacement Data LDA was next applied to the full 3D displacement vectors. The data were projected onto the discriminant axis and visualised in 2D with a linear decision boundary, as shown in Figure 12a. The corresponding discrimination plane in the original 3D space is shown in Figure 12b. The resulting classification accuracy was 90.48%.

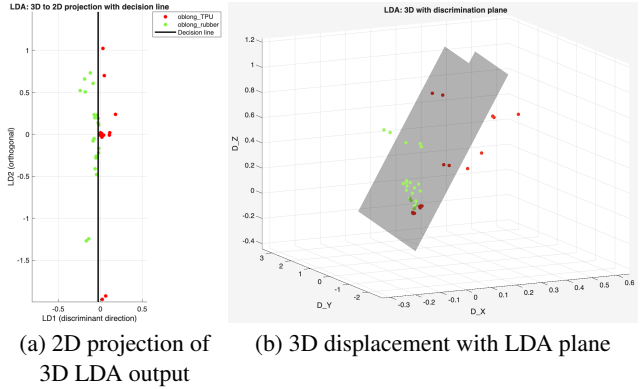


Figure 12: **[D.1.d]** LDA results: (a) 2D projection of 3D output with decision boundary, and (b) 3D central papilla displacement with LDA discrimination plane.

[D.1.e] Physical Interpretation Rubber exhibits greater displacement variance due to increased compliance and damping, while TPU responses are more compact and repeatable. Displacement components aligned with tangential shear (D_X) provide stronger material discrimination than normal-aligned components, explaining the higher accuracies achieved in LDA combinations involving D_X .

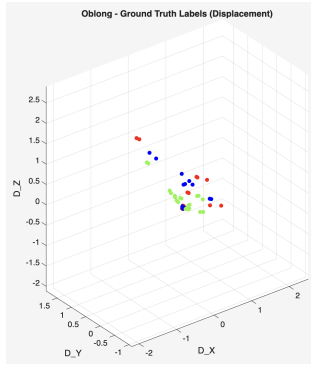
E. Clustering

[E.1.a] Oblong Objects – Central Papilla Displacement Clustering was performed on displacement data from the central papilla (P4) for the oblong objects. Figure 13a shows the raw displacement measurements projected onto the (D_X, D_Y) plane, with points colour-coded by material (TPU, normal, rubber).

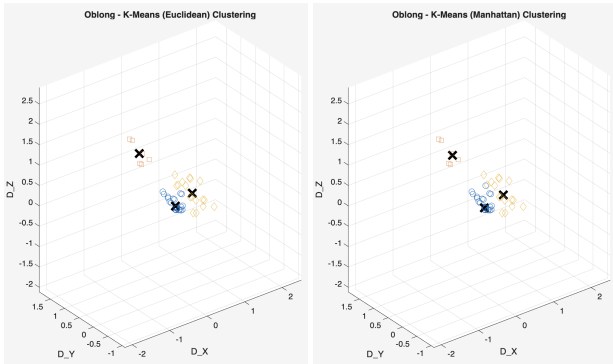
[E.1.b] K-Means Clustering with Euclidean Distance

K-means clustering with $k = 3$ was applied using the squared Euclidean distance metric. Cluster assignments are visualised in Figure 13b, where marker shape indicates cluster membership and colour indicates ground-truth material. The resulting clusters show weak correspondence to material labels, with a measured purity of 36.84%, indicating significant overlap between material displacement responses.

[E.1.c] Effect of Distance Metric The clustering was repeated using the Manhattan (cityblock) distance metric, with results shown in Figure 13c. The resulting cluster purity was 34.92%, compared to 36.51% using Euclidean distance. The similar purity values indicate that changing the distance metric does not substantially improve unsupervised material separation. This suggests that central papilla displacement alone does not provide strong clustering structure for material discrimination.



(a) Ground truth



(b) K-means (Euclidean distance) (c) K-means (Manhattan distance)

Figure 13: [E.1] Central papilla (P4) displacement for oblong objects: ground truth and K-means clustering results under different distance metrics.

F. Gaussian Mixture Model

[F.1.a] 2D Displacement Scatter (Oblong, P4) The left figure 16 shows a 2D scatter plot of central papilla (P4) displacement data for oblong objects, using (D_X, D_Z) as axes.

[F.1.b] Gaussian Mixture Model Contours A Gaussian Mixture Model (GMM) with three components was fitted to the displacement data. Figure 14 shows the resulting probability density contours overlaid on the original scatter plot.

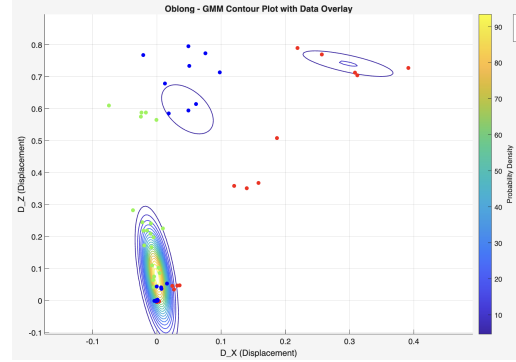


Figure 14: [F.1.b] GMM probability density contours (3 components) overlaid on oblong P4 displacement data.

[F.1.c] GMM 3D Probability Density Surface The fitted GMM density is visualised as a 3D surface in Figure 15. An isometric viewing angle is used to clearly show all mixture components and their relative contributions.

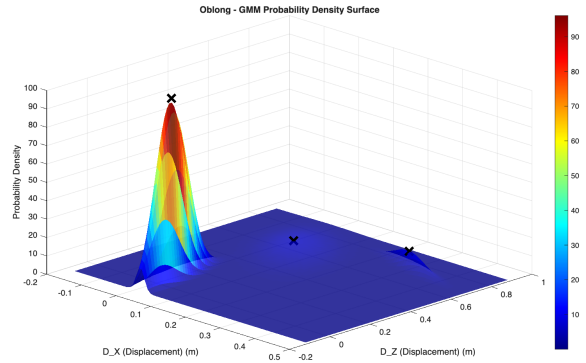


Figure 15: [F.1.c] 3D surface plot of the fitted GMM probability density for oblong P4 displacement data.

[F.1.d] Hard Cluster Assignments Each data point was assigned to the most likely Gaussian component. Figure 16 compares the ground-truth material labels with the hard GMM cluster assignments.

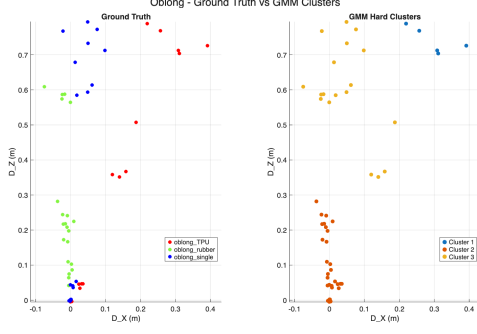


Figure 16: [F.1.a & F.1.d] Hard cluster assignments from the 3-component GMM applied to oblong P4 displacement.

[F.1.e] Comparison to Ground Truth The GMM produces partial alignment with the ground truth labels. One component captures a pure subset of TPU samples, while the dominant component (63.47% mixture weight) contains a mixture of all three materials. The overall clustering purity of 47.62% shows that shear displacement contains some material-specific structure, but is insufficient for complete unsupervised separation. Shear displacement alone therefore provides limited discriminative information, as different materials can produce similar displacement responses during contact.

G. Bagging

[G.1.a] Number of Trees / Bags and Justification A bagged ensemble of 50 decision trees was used, balancing performance and computational cost. Displacement data from all nine papillae (27 features) were reduced via PCA to 6 principal components explaining 95% of the variance.

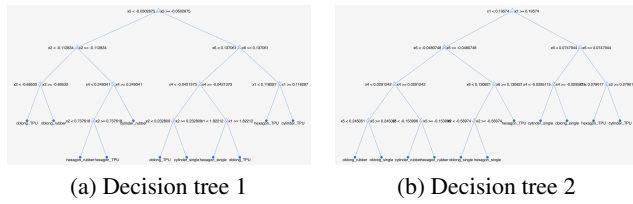


Figure 17: [G.1.b] Two representative decision trees from the bagged ensemble trained on PCA-reduced displacement features.

[G.1.b] Visualisation of Decision Trees Two representative decision trees from the trained bagged ensemble are shown in Figure 17. The trees differ in structure due to bootstrap sampling, illustrating how bagging introduces model diversity and reduces overfitting by averaging across multiple decision boundaries.

[G.1.c] Confusion Matrix and Classification Accuracy

The confusion matrix is shown in Figure 18. The overall test accuracy was 56.76%. All cylinder classes and hexagon rubber were classified perfectly, indicating strong separability for rigid objects. Performance was lower for oblong classes and softer materials due to greater overlap in displacement responses. The out-of-bag error converged to 46.36%, indicating moderate generalisation.

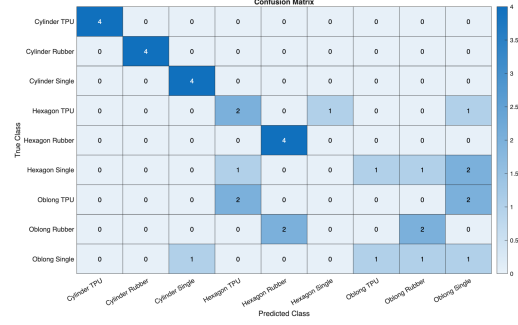


Figure 18: [G.1.c] Confusion matrix for the bagged decision tree classifier. Classes correspond to object shape and material.

[G.1.d] Discussion of Misclassifications and the PCA Step

Misclassifications occur mainly among oblong objects and between softer materials (TPU and rubber), which exhibit similar compliance during contact. Variation in contact location further increases overlap. PCA was beneficial, reducing 27 features to 6 components while retaining 95% variance, though subtle low-variance material cues may have been lost. However, subtle material-specific features contained in lower-variance components may still be lost, particularly for geometrically complex shapes such as oblong objects.

H. Conclusion

[H.1.a] Treating Contacts as Distinct Rather than Identical

The current approach treats all contacts as identical. Performance could improve by conditioning analysis on contact context (e.g., force, angle, or papilla activation), enabling more physically consistent comparisons. This may enhance interpretability and classification accuracy, but increases complexity and reduces data per group, raising overfitting risk.

[H.1.b] Incorporating Temporal Information from the Time Series

The analysis uses a single time step per contact, ignoring temporal dynamics. Including time-based features or full sequence modelling could capture loading and micro-slip behaviour relevant to material discrimination. However, this adds computational complexity and requires careful feature engineering and alignment.

Appendix

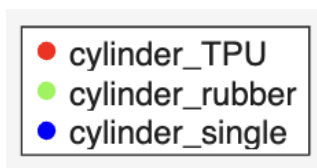


Figure 19: Material to colour labels